ANNUAL REPORT
2024-2025

THE AUTOMATION AND ROBOTICS SOCIETY

TABLE OF CONTENTS

INTRODUCTION

The Automation and Robotics Society (**TARS**) of IIIT Bhubaneswar is a student-driven technical society dedicated to promoting innovation, research, and practical learning in the fields of automation, robotics, and allied technologies. The society aims to cultivate a culture of technical curiosity, teamwork, and problem-solving among students through structured activities and collaborative projects.

TARS comprises 35 active members, currently guided by the Secretary Jagdeswar Pati and two Joint Secretaries Ankit Muduli and Shruti Patro, who oversee the functioning of the society across four key domains — **Technical**, **Content & Research**, **Media & Graphics**, and **Marketing & Outreach**.

- **Tech:** The Technical Domain focuses on the design, development, and implementation of hardware and software-based projects that embody the principles of automation and intelligent systems.
- **Content & Research:** The Content & Research Domain emphasizes technical writing, documentation, research papers and exploration of emerging trends in robotics and automation.
- **Media & Graphics:** The Media & Graphics Domain manages the society's digital presence, visual communication, and event promotion.
- **Marketing & Outreach:** The Marketing & Outreach Domain undertakes the responsibility of external collaboration, sponsorship, and engagement with other technical communities and organizations.

Throughout the academic year guided by Former Secretary Abhijit Mukharjee, and Joint Secretaries Pradosh Preetom and Priyanshu Dwibedi the society has organized several initiatives, including workshops, technical sessions, competitions, and project showcases, all aimed at enhancing students' technical proficiency and practical exposure. These activities have contributed to the institute's broader objective of fostering experiential learning and interdisciplinary collaboration.

This **Annual Report** presents a comprehensive overview of the society's undertakings, achievements, and contributions during the year. It reflects the collective efforts of the members and the leadership in furthering the mission of TARS - to serve as a platform for innovation, learning, and excellence in automation and robotics at IIIT Bhubaneswar.

ORIENTATION 2024

Date: 11th November 2024
TIME: 5:30 PM
VENUE: AG01
ORGANISED BY: TARS – The Automation and Robotic Society

Objective of the Event:

The orientation aimed to introduce freshers to TARS, its vision and mission, and the opportunities in automation, robotics, and innovation. It encouraged students to join and take part in future projects, workshops, and competitions.

Overview:

The orientation session began at 5:30 PM with a warm welcome to all attendees. Core team members and seniors interacted with freshers, sharing insights into TARS's history, domains, achievements, and future plans. The atmosphere was energetic, promoting collaboration and innovation.

Key Highlights:

- **Introduction to TARS:** Covered the society's purpose, structure, and milestones.
- **Project Showcase:** Seniors presented impactful robotics and automation projects.
- **Event Calendar:** Shared details of upcoming workshops, hackathons, and collaborations.
- **Membership Info:** Explained joining procedures and mentorship opportunities.
- **Q&A Session:** Addressed queries on learning resources and involvement.

Engagement & Outcomes:

278+ students participated enthusiastically, with interactive discussions and demos.
Over 150 students showed interest in joining, boosting awareness of robotics and strengthening bonds between seniors and freshers.

Feedback

The orientation received positive feedback, with attendees appreciating:

- The clarity and quality of presentations.
- Inspiring project showcases.
- Exciting games and rewards.
- The welcoming and friendly approach of the core team.

Conclusion

The TARS Orientation successfully introduced freshers to the world of robotics and automation. With motivated students onboard, the society looks forward to a year full of innovation, teamwork, and impactful projects.

INTRODUCTION TO ARDUINO:

Name of the Event- Introduction To Arduino

Date of the Event- 2 December 2024

Location of the Event- CG05, IIIT BHUBANESWAR

Event Summary:

The event took place on 2nd December from 6:30 PM to 8:00 PM at CG05, conducted by Jagdeswar Pati,Former TECH Lead of TARS. The objective was to familiarize students with Arduino and its components. The session began with an interactive quiz testing basic electronics and programming knowledge, serving as an engaging icebreaker and setting a highly interactive tone for the session.

Key Highlights

- **Introduction to Arduino:** Basics of Arduino and its applications in innovative projects were explained.
- **Hands-on Demonstration:** Participants learned to set up the Arduino IDE and explored simple circuits like blinking LEDs and sensor-based projects.
- **Importance of Microcontrollers:** The session emphasized the role of microcontrollers and microprocessors in automation and IoT, inspiring innovation using open-source platforms.
- **Showcase of Sakhha:** The humanoid robot Sakhha was unveiled, demonstrating AI-driven capabilities, natural language processing, human-like interactions, and applications in household, healthcare, and customer service.
- **Interactive Q&A:** Participants' queries were addressed, providing clarity and deeper understanding.
- **Project Opportunities:** Attendees were encouraged to work on real-life projects, and guidance on resources for continued learning was shared.

Feedback & Outcome

The session received overwhelmingly positive feedback, with attendees appreciating its interactivity, hands-on approach, and insights into Arduino and robotics. It was a highly engaging and enriching experience for beginners and enthusiasts alike.

D3: TECHNO-MANAGEMENT FEST

BUILD A BOT: AUTOMATE, EXPLORE, CONQUER!

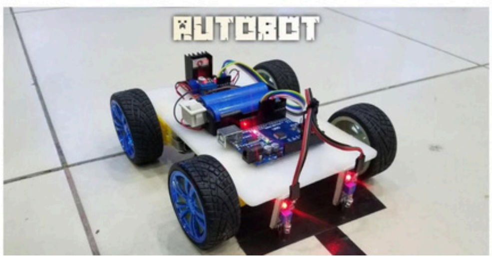
DATE OF EVENT: 8 NOVEMBER 2024
TIME OF EVENT: 10:00 PM - 5:00 PM (Next Day)
EVENT LOCATION: PLACEMENT BUILDING OF IIIT BHUBANESWAR
NUMBER OF TEAM REGISTERED: 27

EVENT SUMMARY:
TARS of IIIT Bhubaneswar hosted a national-level robotics competition on 8th November 2024, attracting 27 teams from across India and generating 43K+ online impressions. Teams of up to 4 members had 7 hours to design and showcase robots, with AI tools allowed. Six teams qualified for the finals, and the winning teams, chosen for fastest completion and minimum penalties, received a prize pool of 30k.

AUTO BOT:

DATE OF EVENT: 9 NOVEMBER 2024
TIME OF EVENT: 10:00 AM- 6:00 PM
EVENT LOCATION: A BLOCK OPEN SPACE
NUMBER OF TEAMS REGISTERED: 15

EVENT SUMMARY:
The Auto Bot event, a highlight of D3 Fest, showcased participants' innovation in autonomous robotics. Teams designed and programmed robots to navigate predefined tracks while tackling twists, turns, and obstacles with speed and precision.

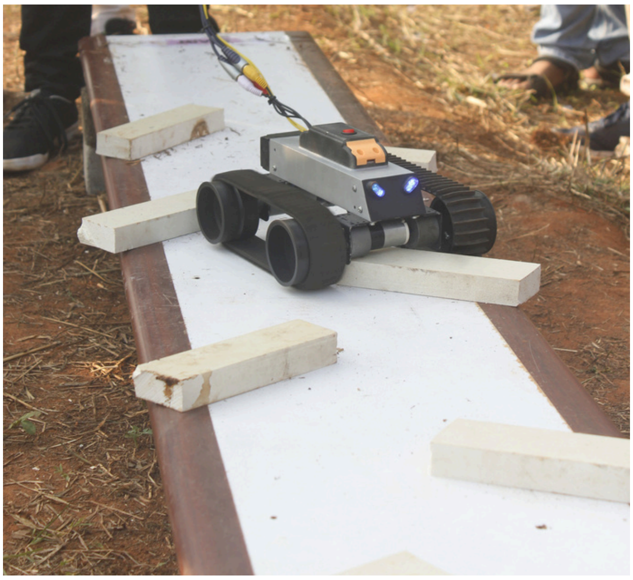


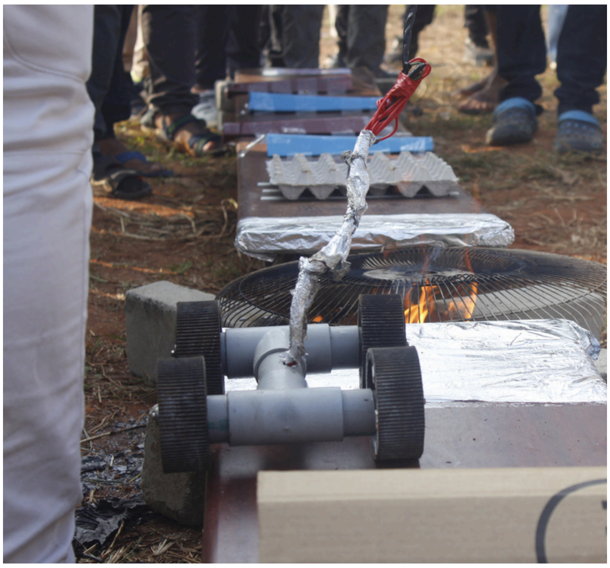
The competition tested expertise in sensors, programming, and control systems, pushing the boundaries of automation. The event attracted 15 teams from across India, with 5 qualifying for the finals, and garnered over 14K online impressions. The top 2 teams were awarded a total cash prize of 10k.

ROBO RACE:

DATE OF EVENT: 9 NOVEMBER 2024
TIME OF EVENT: 10:00 AM- 6:00 PM
EVENT LOCATION: OPEN FIELD
NUMBER OF TEAMS REGISTERED: 35

EVENT SUMMARY:
As part of D3 Fest, the Robo Race took place on 9th November 2024 at the Open Field, bringing together 35 teams in an exciting robotics competition. Participants navigated obstacles, showcasing engineering skills, creativity, and optimization for speed and efficiency. The event attracted over 20k online impressions and highlighted innovation, collaboration, and problem-solving in robotics. The thrilling race combined technology and competition, inspiring attendees and celebrating the practical applications of robotics in real-world scenarios. The top 2 teams were awarded a total cash prize of 15k.





ADVAITA 2K25: TECHNO-CULTURAL FEST

EVENT : INVENTO EXPO

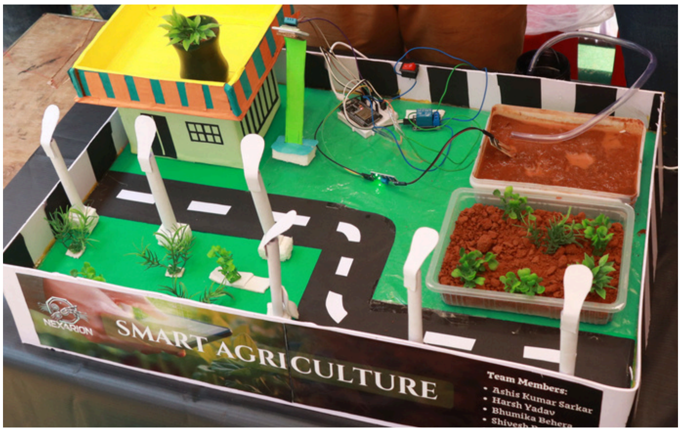
DATE OF EVENT: 29th - 30th MARCH 2025
TIME OF EVENT: 11:00 AM- 6:00 PM
EVENT LOCATION: A BLOCK OPEN SPACE
NUMBER OF TEAMS REGISTERED: 30

EVENT SUMMARY:
The Invento Expo, part of Advaita 2025, was held on 29th–30th March at A Block Open Space, IIIT Bhubaneswar, showcasing student-led innovations and blending creativity, technology, and entrepreneurship.

OVERVIEW
The event began with an online PPT submission round, where teams presented project ideas, implementation strategies, and potential impact. The top 15 teams were shortlisted for the offline exhibition round.

EXHIBITION HIGHLIGHTS
Over two days, teams displayed MVPs, prototypes, and hardware solutions to 200–300 visitors, including industry leaders, investors, researchers, faculty, and technology enthusiasts. Attendees engaged in discussions about real-world applicability and scalability. The Expo also generated 30k+ online impressions, amplifying its reach.

SIGNIFICANCE
Celebrating innovation and engineering excellence, Invento Expo provided a platform for students to transform ideas into market-ready solutions, bridging academic creativity with real-world applications and inspiring future-focused technologies.



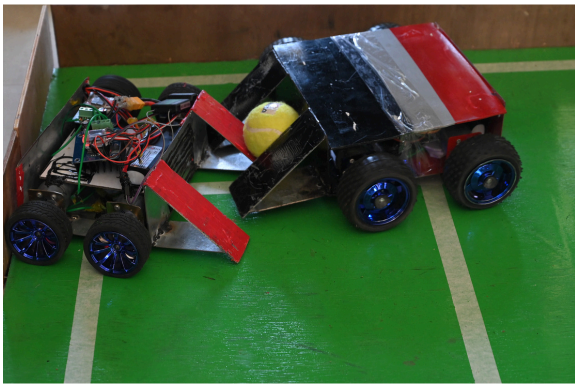
ROBOROGUE (ROBO SOCCER)

DATE OF EVENT: 29th MARCH 2025
TIME OF EVENT: 10:00 AM- 6:00 PM
EVENT LOCATION: C BLOCK OPEN SPACE
NUMBER OF TEAMS IN OFFLINE ROUND : 14

EVENT SUMMARY:
As part of Advaita 2025, RoboRogue took place on 29th March at C Block Open Space, IIIT Bhubaneswar, bringing the excitement of soccer to robotics. The event featured 14 teams with on-spot registrations, where custom-built bots demonstrated strategy, speed, and precision in a high-energy robotic soccer competition.

HIGHLIGHTS
Robots dribbled, passed, and defended using both manual and autonomous approaches, captivating a crowd of 150–200 spectators. The event emphasized teamwork, innovation, and quick thinking, creating a spirited competitive environment.

IMPACT
RoboRogue generated over 20k+ online impressions, showcasing the blend of sportsmanship and technology and inspiring participants and audiences alike.

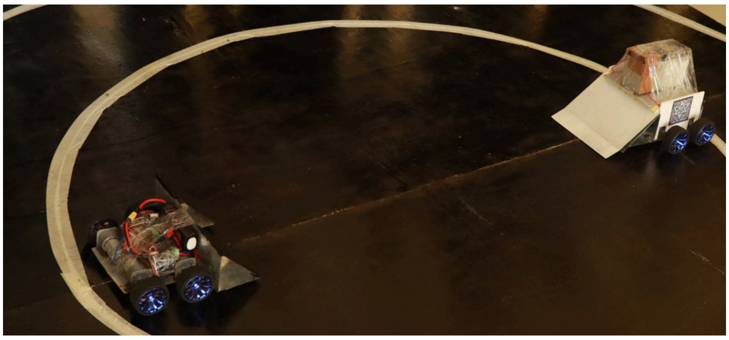
BRAWLBOUNTY (ROBO SUMO)

DATE OF EVENT: 28th MARCH 2025
TIME OF EVENT: 10:00 AM- 6:00 PM
EVENT LOCATION: C BLOCK OPEN SPACE
NUMBER OF TEAMS IN OFFLINE ROUND : 10

EVENT SUMMARY:
Advaita 2025 began with BrawlBounty, a thrilling bot battle, on 28th March at C Block Open Space, IIIT Bhubaneswar. Teams registered on-site and competed with combat-ready bots in double-ring knockout battles, where the inner ring awarded points for strategic pushes and the outer ring offered instant victory.

HIGHLIGHTS
Matches progressed through quarter-finals, semi-finals, and a grand finale, keeping excitement high. Around 200+ spectators cheered as bots showcased power, precision, and tactical maneuvers.

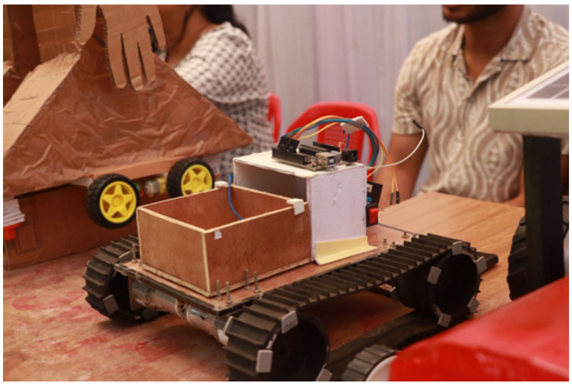
IMPACT
With strict rules ensuring fair play, BrawlBounty highlighted engineering skill, driver control, and strategy, creating an electrifying start to Advaita 2025, having 21k+ impressions.

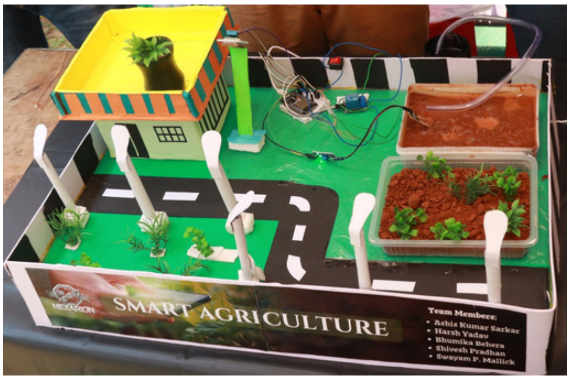


COMPLETED PROJECTS

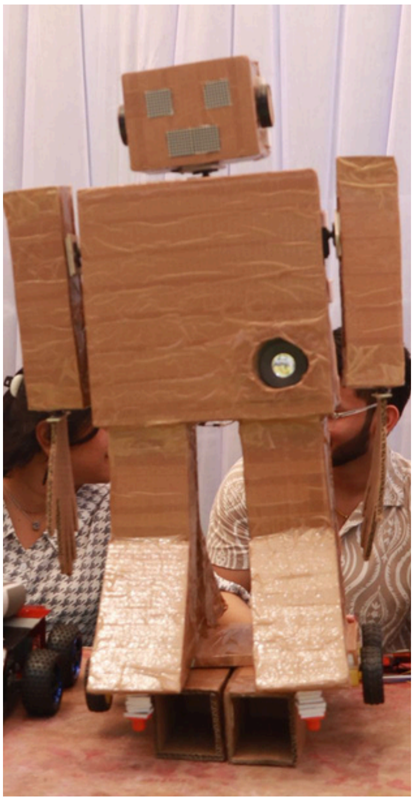
Hardware Projects

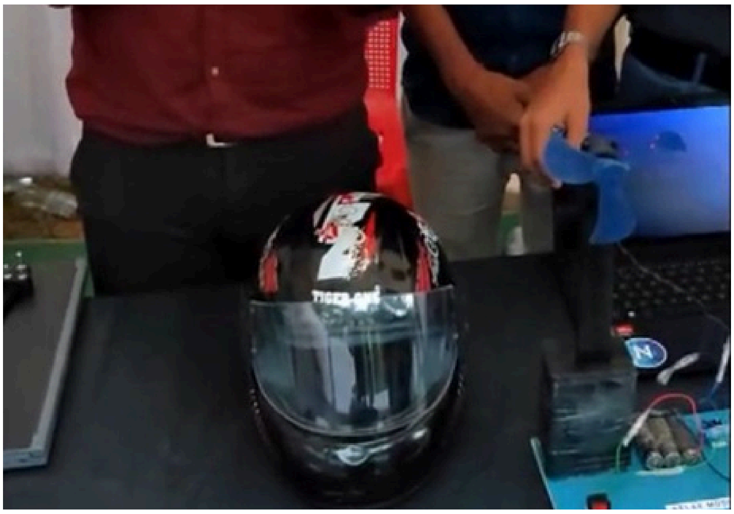
1. **Smart Water Segregation System:** Designed an automated system capable of distinguishing and separating clean and waste water using sensors and microcontrollers to promote sustainable water management.
2. **IoT-Based Smart Home Management System:** Developed a smart home setup that enables users to control lighting, appliances, and security remotely through IoT integration and mobile app connectivity.
3. **Smart Agriculture Model:** Created an intelligent farming prototype using sensors for soil moisture, temperature, and humidity to optimize irrigation and crop health monitoring.
4. **Smart Helmet:** Built a safety helmet equipped with accident detection and alcohol-sensing features to ensure rider safety and alert emergency contacts instantly.
5. **Robo Soccer and Robo Sumo Bot:** Constructed competitive robots designed for strategic movement and precision control to participate in Robo Soccer and Sumo robotics events.
6. **Sand Rover Bot:** Engineered an all-terrain rover capable of navigating through sand and uneven surfaces, emphasizing mechanical design and stability control.
7. **Water Rocket Project:** Designed and launched a high-performance water-propelled rocket demonstrating aerodynamics, fluid dynamics, and thrust optimization.

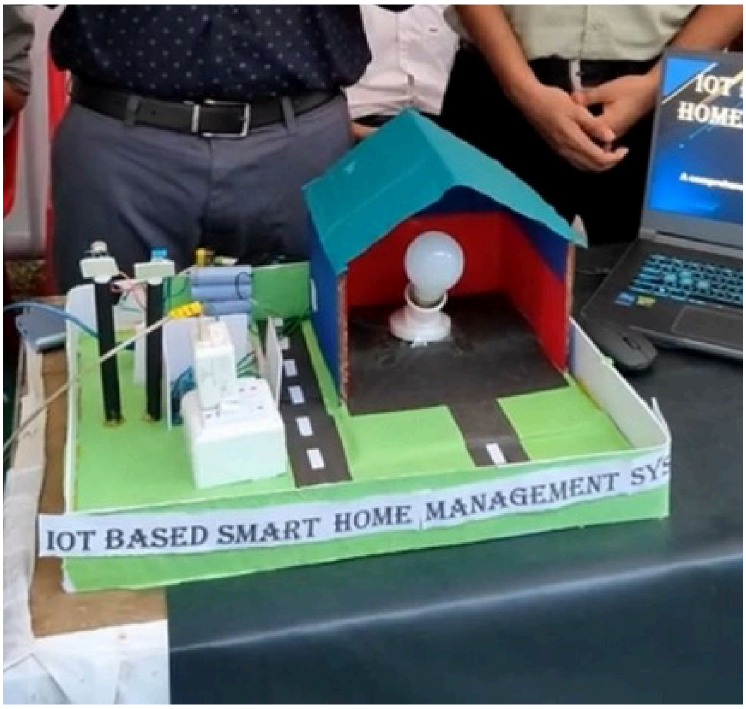




7. **Humanoid Robot "Sakhha" Prototype:** Developed the prototype of a semi-autonomous humanoid robot capable of performing basic movements and human-like interactions.
8. **Line Follower Bot:** Created a sensor-based robot that detects and follows a predefined path, showcasing precision control and efficient algorithmic design.

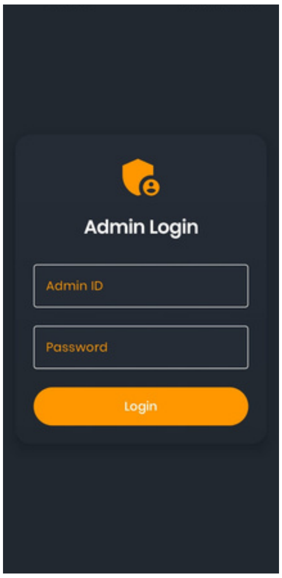


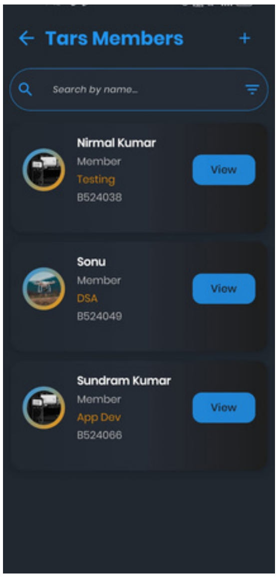


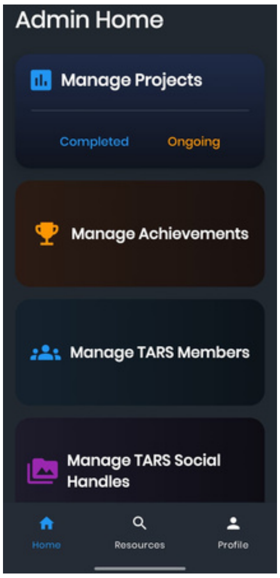


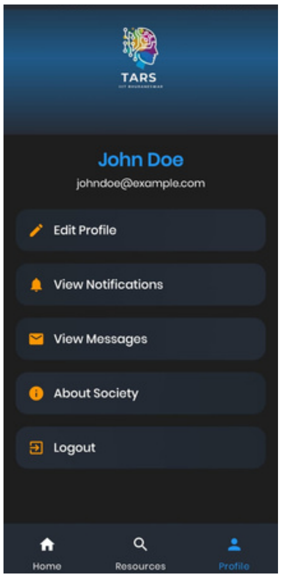
Software Projects

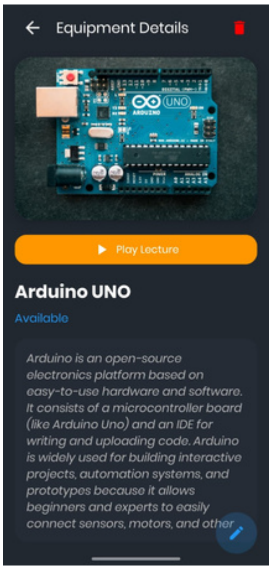
- **TARS Website:** Developed and deployed an official team website to showcase projects, achievements, and research initiatives, enhancing TARS's digital presence.
- **Sakhha Desktop Model:** Built a desktop-based interface to simulate and control the humanoid robot "Sakhha," integrating real-time commands with AI-based automation features.











Sakhha Desktop Model UI

ONGOING PROJECTS

Hardware Projects

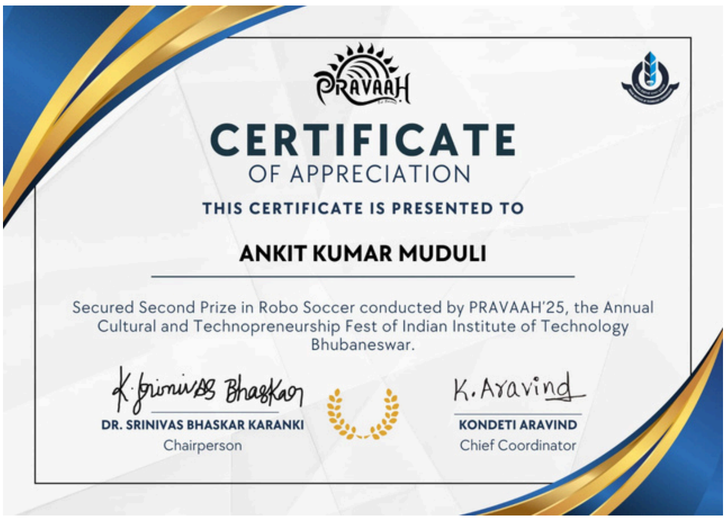
1. **Humanoid Robot – Complete Working Model:** An advanced version of the "Sakhha" humanoid, this project aims to develop a fully functional robot capable of performing human-like tasks, gestures, and communication using AI and embedded systems.
2. **Assistant-Based RC Plane:** A remote-controlled aircraft integrated with an onboard voice assistant for navigation and data communication, combining aerodynamics with AI-based automation.
3. **Gesture Control Bot:** A robotic system being developed to respond to hand gestures using accelerometer and gyroscope sensors, enabling intuitive and contactless robot control.
4. **Small Learning Projects:** A series of experimental builds focused on learning and testing new technologies, components, and algorithms for robotics and embedded systems.
5. **Agriculture Bot:** A semi-autonomous robot under development to assist farmers in ploughing, seeding, and soil analysis using sensor-driven precision farming techniques.

Software Projects

1. **TARS Cloud App:** A web-based cloud platform in progress to manage project data, automate report generation, and facilitate remote team collaboration.
2. **TARS Inventory App:** An application being designed to streamline component tracking, borrowing logs, and real-time inventory management within the TARS lab.
3. **Sakhha Model for Lab Automation:** A smart software system being developed to integrate the humanoid robot "Sakhha" with lab devices, enabling automated control and monitoring of experiments.

MILESTONES:

- SECURED **2ND** PRIZE IN THE WATER ROCKET PROJECT AT **VSSUT, BURLA (ODISHA)**.
- EMERGED AS THE **WINNER** OF THE BUILD A BOT COMPETITION IN D CUBE, HELD AT **IIIT BHUBANESWAR**.
- RANKED AMONG THE **TOP 5** IN THE SAND ROVER EVENT AT KSHITIJ TECH FEST, **IIT KHARAGPUR**.
- ACHIEVED **1ST** PRIZE IN ROBO ROGUE AT ADVAITA, IIIT BHUBANESWAR.
- SECURED **2ND** PRIZE IN BRAWL AND BOUNTY AT **ADVAITA, IIIT BHUBANESWAR**.
- WON **2ND** PRIZE IN ROBO SOCCER AT PRAVAAH’25, **IIT BHUBANESWAR**.


PRAVAAH

CERTIFICATE
OF APPRECIATION

THIS CERTIFICATE IS PRESENTED TO

ANKIT KUMAR MUDULI

Secured Second Prize in Robo Soccer conducted by PRAVAAH'25, the Annual Cultural and Technopreneurship Fest of Indian Institute of Technology Bhubaneswar.

DR. SRINIVAS BHASKAR KARANKI
Chairperson

KONDETI ARAVIND
Chief Coordinator

SUMMARY:

The Automation and Robotics Society (TARS) of IIIT Bhubaneswar is a student-led body promoting innovation, research, and hands-on learning in automation and robotics. With **35** active members across four domains-**Tech**, **Content & Research**, **Media & Graphics**, and **Marketing & Outreach** - TARS undertook several impactful initiatives during 2024-25.

KEY EVENTS & ACTIVITIES

- **Orientation (Nov 2024):** Introduced 278+ students to TARS, boosting awareness and participation.
- **Workshops:** Hands-on Arduino and sensor sessions built strong foundations in IoT and automation.
- **D³ Tech Fest (Nov 2024):** National-level robotics contests (Build-a-Bot, Robo Race, Auto Bot) attracted 27+ teams and 43,000+ online engagements.
- **Advaita 2025 (Mar 2025):** Invento Expo, RoboRogue, and BrawlBounty showcased student innovations to 200-300 visitors, including industry experts.

PROJECTS & INNOVATION

Through a structured Arduino lab series, teams developed prototypes addressing real-world needs, including:

- Smart Helmet for safety
- Smart Dustbin for hygiene
- Home Automation solutions
- Smart Irrigation and environmental monitoring systems and many more.

OUTREACH & ACHIEVEMENTS

TARS members participated in events at IIT Kharagpur, Silicon Institute, and others, while expanding visibility through LinkedIn and Instagram.

CONCLUSION

The year 2024-25 marked significant growth for TARS in technical activities, student engagement, and external outreach-strengthening IIIT Bhubaneswar's presence in the field of robotics and automation.

THANK YOU

The Automation and Robotics Society (TARS) of IIIT Bhubaneswar extends its sincere appreciation to all mentors, faculty members, collaborators, and students for their constant support and contribution throughout the academic year.
Together, we continue to foster a culture of innovation, teamwork, and technical excellence in the fields of automation and robotics.

“Innovation is born when curiosity meets collaboration.”

IIIT Bhubaneswar | The Automation and Robotics Society (TARS)

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